# WEIYI HUANG

(512) 902-6038 wh474@cornell.edu linkedin.com/in/wh474

## EDUCATION

**Cornell University,** College of Engineering, Bachelor of Science **Expected May 2027**
Electrical and Computer Engineering, Aerospace Engineering Minor
***Relevant Courses:*** Computer Architecture, Circuits, Microelectronics, Probability for Random Signals, Embedded Systems (SP25), Thermodynamics, Statics and Mechanics of Solids, Fluid Mechanics (SP25), Spaceflight Mechanics (SP25)

**Phillips Exeter Academy,** High School, Exeter, NH **Aug 2020-Jun 2023**

## ENGINEERING EXPERIENCE

**MAVRIK, Inc,** *Lead Electrical Engineer* **June 2025-Present**

- *Worked on entire product line: HATCHET (Group 3 hybrid UAS w/ 500-lb payload), CHORUS, Skimmer, and Dart-M*
- Prototyped embedded hardware PCB with KiCad to support modular payload release mechanisms (servos and latches), interfacing with STM32 and Cortex-M7-based microcontrollers through NVIDIA Jetson Nano companion computer.
- Developed low-level C++ code to interface payload board with custom companion software running Linux.
- Architected a redundant WAN failover solution using Starlink, DoodleLabs, and LTE through an AWS Lightsail instance to ensure persistent communication between GS and onboard companion computer.
- Implemented all flight-critical communication from Cube Orange, such as ESCs (CAN/PWM), uBlox GPS (I2C + UART), etc. Customized base set of MissionPlanner parameters to support rapid deployment of 10+ drones.
- *Spearheaded electrical efforts and system level integration to win US Army's xTech Overwatch event in October 2025.*

**Cornell University AutoDrone,** *Electrical Subteam Lead* **Sept 2024-Present**

- Directed the implementation of all electrical systems for the high-speed "LawnDart" UAS, including RX/TX, FPV, and a high-discharge power architecture; performed extensive troubleshooting of electrical systems and INAV firmware to achieve reliable flight tests and consistent benchmarks for 250+ mph world record attempts.
- Engineered a custom motor test stand using an Arduino-based embedded system and supporting sensors to conduct rigorous performance evaluations, measuring stress force, RPM, and EMI noise under maximum throttle conditions.
- Appointed as Electrical Subteam Lead to lead architecture of 6+ UAS drones, based on prior success in reliable design.

**Cornell University Biomedical Device (CUBMD),** *Product Development and Media Subteam Member* **Feb-Dec 2024**

- Developed a wireless pump control prototype using an Arduino-based embedded solution, enabling app integration via Bluetooth connectivity with the HC-08 Bluetooth module; iterated on prototype design by engineering a custom PCB to integrate all components, supporting an ATiny85 microcontroller in a condensed form factor.
- Generated 5,000+ new impressions and increased team applications by +25% through streamlining website redesign.

## ENGINEERING EXPERIENCE

**Multi-Core RISC-V Processor,** *(Personal Project; Computer Architecture Final Project)* **August 2024-December 2024**

- Developed hardware and software for multi-core RISC-V architecture, creating custom ring network interconnect with dedicated routing and arbitration units to support thread-level parallelism through multi-threaded C programs.

**FET Amplifier Design and Optimization,** *(Personal Project; Microelectronics Final Project)* **January 2025-June 2025**

- Designed and constructed a cascaded FET amplifier and cascode amplifier using ALD1105 chips; characterized amplifier performance by measuring voltage gain, output resistance, and maximum voltage swing to ensure transistors remained in the saturation region and minimized distortion of input signal.

## LEADERSHIP EXPERIENCE

**Project Level the Field (Project LTF),** *Chapter Founder and Vice President* **May 2024-Present**

- Founded Project LTF's Cornell chapter, a 501(c)(3) nonprofit providing college app resources to underserved students.
- Collaborated with a 7-member executive board, 11 mentors, and national chapter leadership to develop a 12-week strategic blueprint and execute 2 workshops, effectively sharing college app resources to 60+ underprivileged students.

## SPECIALIZED SKILLS

***Embedded Systems and Avionics****:* C/C++, Python, Verilog/SystemVerilog, Bash scripting; STM32, Cortex-M7, ATtiny, Arduino, Raspberry Pi; avionics buses (CAN, UART, SPI, I2C, PWM), sensor/actuator integration, EMI mitigation, redundant and fault-tolerant power architectures, flight control electronics, telemetry and data links, motor controllers.
***Electronics & Tools****:* KiCad, LTSpice/SPICE simulation, Quartus Prime; oscilloscopes, spectrum/logic analyzers, DMM; soldering, harness fabrication, rapid prototyping, EMI testing; CAD/FEA (SolidWorks, Onshape, ANSYS); Linux, Git/GitHub.